

Learning Activity Sheet for General Mathematics

Quarter 10

Unit

10

Learning Activity Sheet for Mathematics Grade 11
Quarter 4: Unit 10

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LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	4
Lesson No.:	10.1	Date:	
Lesson Title/ Topic:	Problem Solving Involving Compound Interest		
Name:		Grade & Section:	

I. Activity No. 1	Practicing Learned Skills
II. Objectives:	At the end of the activity, you should be able to solve problems involving compound interest with varying compound periods.
III. Materials Needed:	pen and paper
IV. Instructions:	Solve the following problems by following the given example. 1. Vincent is saving up to buy a motorcycle worth ₱50,000 in 5 years. He plans to invest her money in an e -wallet Save account that offers 6% interest compounded annually. How much should he deposit today so that he can buy her motorcycle after 5 years? 2. Francis invested 20,000 in Golden Peso Bank, which offers 5% annual interest compounded annually. He wants his money to grow to 30,000. For how many years should Francis keep his money in the bank to reach 30,000? 3. Racquel invested 35,000 in two different banks for 4 years, compounded annually: *Golden Peso Bank: After 4 years, her money grew to P45,000. *Mabuhay Savings Bank: After 4 years, her money grew to P47,000. Which bank offered a higher annual interest rate?
V. Synthesis:	1. Which of the three problems requires using the present value, which uses the time period, and which requires comparing interest rates? Explain how you determined which variable to solve for in each case. 2. If you were saving for a major purchase, which strategy would you prefer: depositing a larger amount now, waiting longer with a smaller amount, or choosing a bank with a higher interest rate? Explain your choice using examples from the problems. 3. Suppose you have ₱35,000 and want to reach ₱50,000 in 5 years. Based on the interest rates found in the problems, which bank would you choose and why? Show the calculation to support your decision

I. Activity No. 2	Making Real-World Connections
II. Objectives:	At the end of the activity, you should be able to solve problems involving compound interest with varying compound periods.
III. Materials Needed:	pen and paper
IV. Instructions:	Pick one problem (written on the strip of paper) and solve it by group. Discuss your answer and solution with the group and present your output. La Union Cooperative Loan

	A fisherman in La Union borrowed ₱50,000 from La Union Cooperative at 6% annual interest, compounded semi-annually. How much will he owe after 4? Student Savings for Tuition Marilou wants to have ₱80,000 saved for her college tuition after 5 years. She plans to deposit her money in Golden Peso Bank, which offers 8% compounded annually. How much should she deposit today? Teacher's Pag-IBIG MP2 Investment Mr. Obra, a teacher, invested ₱100,000 in the Pag-IBIG MP2 program, which gives 7% compounded annually. He wants his money to grow to ₱150,000. How many years will it take for his investment to reach ₱150,000? Family Business Fund The Dela Cruz family invested ₱200,000 in a Time Deposit Account for their family business expansion. After 6 years, their savings grew to ₱320,000, compounded annually. What annual interest rate did the bank offer? Overseas Worker's Remittance An OFW in Dubai sends ₱100,000 to his family in the Philippines and advises them to deposit it in Golden Peso Bank with 10% interest compounded annually. After 3 years, the money will be used to renovate their house. How much will the family have after 3 years?
V. Synthesis:	- Which of the scenarios results in the highest amount of growth relative to the initial amount invested or borrowed, and what factors (interest rate, time, compounding frequency) make it the most effective? - If you were in the fisherman's, Marilou's, Mr. Obra's, or the OFW family's situation, which financial decision would you consider the most beneficial and why? Use your calculations to support your choice. - How do the concepts of time, interest rate, and compounding frequency work together to influence the final amount in each scenario? Explain how changing one factor (e.g., increasing the interest rate or extending the time) would affect the outcome in at least two different problems.

I. Activity No. 3	Assessing Learning Outcomes
II. Objectives:	At the end of the activity, you should be able to solve problems involving compound interest with varying compound periods.
III. Materials Needed:	pen and paper
IV. Instructions:	Solve the following problems. Show a complete solution. - Jessa bought a new cellphone worth ₱15,000 using a loan from Golden Peso Bank. The bank charges 10% interest compounded annually for 2 years. How much will she pay after 2 years? - Joy, an OFW, plans to remit ₱150,000 to the Philippines for her family's future needs. She has two investment options for 5 years: Option A (Bank Time Deposit): 5% annual compounding Option B (Barangay Cooperative): 2.5% every 6 months (semi-annual compounding) Which

V. Synthesis:	1. Which investment option (Option A or Option B) yields a higher final amount, and what causes the difference in the results even though both options have similar interest rates? 2. If Joy needs the money sooner than 5 years, how might this affect her choice between the two investment options? Explain your reasoning using the concepts of compounding frequency and time. 3. Suppose Jessa wants to avoid paying interest by saving instead of borrowing. How much would she need to save monthly for 2 years to reach ₱15,000, assuming she can earn 10% interest compounded annually? (You may use the future value of annuity formula.)
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LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	4
Lesson No.:	10.2	Date:	
Lesson Title/ Topic:	Simple and General Annuities		
Name:		Grade & Section:	

I. Activity No. 1	Practicing Learned Skills												
II. Objectives:	At the end of the activity, you should be able to solve problems involving annuities.												
III. Materials Needed:	pen and paper												
IV. Instructions:	Activity B.3: Practicing Learned Skills Instructions: Identify the type of annuity and solve the problems below: - Monthly payments of ₱ 2,000 for 5 years with interest rate of 12% compounded annually - Yearly payment of ₱15,000 for 10 years with interest rate of 8% compounded annually. - Mr. Miguel Dimaano, a 50-year old Filipino professional, purchases an annuity from an insurance company to supplement his retirement income. He will contribute a lump sum of P1,500,000. The annual interest rate and the annual payout rate are 5% and 4%, respectively. The annuitization period is 10 years. - Ms. Susan Santos, a 40-year-old marketing executive, purchases insurance from an insurance provider. She decides to make varying contributions during the 15-year accumulation phase. <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th><i>Year (k)</i></th><th><i>Premium Amount (PV)</i></th></tr> </thead> <tbody> <tr> <td>1</td><td>P40,000</td></tr> <tr> <td>2</td><td>P50,000</td></tr> <tr> <td>3</td><td>P50,000</td></tr> <tr> <td>4</td><td>P60,000</td></tr> <tr> <td>5-15</td><td>P0</td></tr> </tbody> </table> The guaranteed annual interest rate is at 4% compounded annually and the assumed payout rate is at 3.5% compounded annually. The annuitization period is 10 years. - What will be the accumulated value at the end of the 15-year accumulation period?	<i>Year (k)</i>	<i>Premium Amount (PV)</i>	1	P40,000	2	P50,000	3	P50,000	4	P60,000	5-15	P0
<i>Year (k)</i>	<i>Premium Amount (PV)</i>												
1	P40,000												
2	P50,000												
3	P50,000												
4	P60,000												
5-15	P0												

	b. How much is the annual income payment during the 10-year annuitization period?
V. Synthesis:	1. How did you classify the payments in the given situation above change the way you approached the problem? 2. How can understanding annuity types help you make wiser decisions about your future investments or loans?

I. Activity No. 2	Making Real-World Connections
II. Objectives:	At the end of the activity, you should be able to solve problems involving annuities.
III. Materials Needed:	pen and paper
IV. Instructions:	Mechanics: A. Group Work (3-5 members): Each group is assigned one localized scenario. B. Problem Analysis and computation: Let the learners solve the assigned problem applying the concept of annuity. C. Decision-Making: Presentation of output applying the assigned strategy for each scenario. <i>Solve the given problem</i> Scenario 1: The Retirement Growth Plan The Situation: You, at age 25, decide to use the MP2 program to save for a future down payment on a larger house. You plan to make regular, voluntary deposits. The Details (Simplified): • Regular Deposit (R): ₱24,000 at the end of every year. • Term (t): 5 years (The minimum holding period for MP2). • Interest Rate (r): Assume a fixed average dividend rate of 7.5% per year. • Compounding/Crediting: Annually (simplified, as dividends are credited annually). The Task (Simple Annuity Mastery): 1. Why is this a simple annuity? 2. Determine the total amount available after 5 years, including earned dividends. Scenario 2: The First Home Loan The Situation: You are applying for a Pag-IBIG Housing Loan to purchase a condo unit. The loan amount you need is determined by the maximum monthly payment you can afford. The Details: • Loan Amount/Present Value (PV): ₱3,000,000 • Loan Term (t): 30 years.

	• Payment Frequency (k): Monthly (12 payments per year). • Interest Rate (r): 6.5% per year. • Compounding Frequency (m): Quarterly (4 times per year). The Task (General Annuity Mastery): 1. Why is this a General Annuity? 2. Find the monthly interest rate that is financially equivalent to 6.5% compounded quarterly. 3. What is the Present Value in the given problem? Use the present value formula, rearranged to solve for R. Scenario 3. Fair Market Value Integration To meet the third objective (calculating Fair Market Value, FMV), the housing loan scenario can be extended: After 10 years (120 payments), you receive a bonus and decide to pay off the remaining loan. The bank gives you the option to pay a single lump sum today. Calculate the FMV (Fair Market Value) of the remaining cash flow (the last 20 years, or 240 payments). This FMV is your remaining principal balance. Scenario 4: “OFW Retirement” Case Prompt: A returning OFW, Manuel, is 40 years old. He has saved P1,000,000. He plans to use this for retirement, which starts at age 65. He wants to receive a monthly income for 15 years, starting at age 65.
V. Synthesis:	1. How do the simple annuity in Scenario 1 and the general annuity in Scenario 2 differ in terms of how payments and interest are structured, and how does this affect the total amount saved or the monthly payment required? 2. Considering Scenarios 1, 2, and 4, how does time (number of years) influence the final outcome when all other factors are constant? Which scenario shows the greatest benefit of time, and why? 3. If you were in Scenario 3 and had the option to pay the lump sum after 10 years, would you choose to pay off the remaining balance early or continue with the regular payments? Explain your decision using the concepts of FMV, outstanding balance, and interest savings.

I. Activity No. 3	Assessing Learning Outcomes
II. Objectives:	At the end of the activity, you should be able to solve problems involving annuities.
III. Materials Needed:	pen and paper
IV. Instructions:	A. Please read the sentences carefully and fill in the missing word/s by writing your answer on the line/s provided. 1. A sequence of payments made at equal (fixed) intervals or periods of time is called _____. 2. Each payment in an annuity is called _____. 3. The time between the successive payments dates of an annuity is called the _____. 4. In simple annuity the payment interval is also the same as the _____.

	5. In general annuity the interest conversion or compounding period is _____ with the payment interval. B. Solve the following problems. Follow the five-step process. 1. Your mom asked your opinion if she will be joining the cooperative in her office. She wants to know the amount of money she will be receiving after 6 months and wants to buy something in December. The cooperative wants her to contribute ₱1,000 per month beginning in June 2020 which will earn 3% compounded monthly. How much will be the future value of your mom's contribution at the end of December 2020 2. Mr. Jose dela Cruz, a 45-year-old OFW, just received a retirement bonus of P1,500,000. He decides to use this lump sum to purchase SPDA to supplement his retirement income starting at age 65. The insurance offers a guaranteed rate of 4.5% per annum, compounded quarterly, during both deferral and payout phases. He wants his annuity to provide him with monthly retirement income for 15 years. a. What is the accumulated value of the single premium? b. What is the guaranteed monthly payout Mr. dela Cruz will receive starting at age 65? 3. Ms. Ana Reyes, a 45-years old overseas Filipino worker (OFW), decides to use a portion of her savings to purchase an annuity from a Philippine insurance company to ensure an additional retirement income starting at age 60. She will contribute an amount of P1,000,000. The annual interest rate is at 4.5% and the payout rate is at 4%. The guaranteed payout period is 10 years. a. What will be the accumulated value of the single premium at the end of the 15-year deferral period? b. What is the amount of the annual income payment Ms. Reyes will receive for 10 years during the payout period?
V. Synthesis:	1. How do the problems in Part B demonstrate the difference between accumulation (growing the investment) and payout (receiving income), and why is it important to separate these two stages when planning for retirement? 2. If Mr. dela Cruz and Ms. Reyes both want to receive the same monthly/annual income, but one of them wants to start receiving it earlier, how would their required initial investment or payout period change? Explain using the relationship between time, interest rate, and payment amount. 3. Compare the effects of monthly compounding (Problem 1) and quarterly compounding (Problem 2) on the final value. Which compounding frequency results in a higher future value, and why? Use your calculations to support your answer.

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	4
Lesson No.:	10.3	Date:	
Lesson Title/ Topic:	Business and Consumer Loan		
Name:		Grade & Section:	

I. Activity No. 1	Practicing Learned Skills
II. Objectives:	At the end of the activity, you should be able to solve problems involving solve problems involving business and consumer loans.
III. Materials Needed:	pen and paper
IV. Instructions:	Solve the following: - Mr. and Mrs. Banal purchased a house and lot worth P4,000,000. They paid a down payment of P800,000. They plan to amortize the loan of P3,200,000 by paying monthly for 20 years. The interest rate is 12% convertible monthly. - How much is the monthly payment? - What is the total interest paid? - Mrs. Rosal bought a car. After paying the down payment, the amount of the loan is P400,000 with an interest rate of 9% compounded monthly. The term of the loan is 3 years. How much is the monthly payment? - Mrs. Tan got a business loan worth P800,000. She promised to pay the loan semi-annually in 5 years. The semi-annual payment is P103,603.66 if money is worth 10% converted semi-annually. How much is the outstanding balance after the third payment?
V. Synthesis:	- Mr. Banal and Mrs. Rosal both took loans with different interest rates and terms. Based on the monthly payments and total interest paid, which loan is more expensive and why? Hint: Compare total interest relative to loan amount and duration. - Mrs. Tan is considering paying off her loan early after the 3rd payment. Using the outstanding balance computed, should she pay off the remaining balance now or continue paying monthly? Explain your answer based on financial benefits and risks. - If Mr. and Mrs. Banal want to reduce the total interest paid, which strategy would be most effective: - Increasing monthly payments, - Shortening the loan term, or - Negotiating a lower interest rate? Explain your choice and how it affects the total interest.

I. Activity No. 2	Making Real-World Connections
II. Objectives:	At the end of the activity, you should be able to solve problems involving solve problems involving business and consumer loans.
III. Materials Needed:	pen and paper

IV. Instructions:

Below are four activity options. Choose one activity that matches your interest. Then, find classmates who have chosen the same activity to form your group.

Option 1: Role Play

1. Form groups of 5-6 members.
2. Assign roles: one student as the loan officer, one as the client, and others as family/friends.
3. The client presents a need (e.g., buying a house, starting a small business).
4. The loan officer must:
 - Identify if it is a Business Loan or Consumer Loan.
 - Explain the characteristics (purpose, repayment terms, requirements, interest rate).
 - Show a sample computation of a monthly payment.
5. Present your role play to the class in 3–5 minutes.

Option 2: Poster / Infographic

1. Work individually or in pairs.
2. Create a **poster or infographic** comparing **Business Loans vs. Consumer Loans**.
3. Your work must include:
 - Purpose
 - Repayment Terms
 - Requirements
 - Interest Rates
4. Add a solved example problem (e.g., mortgage or consumer loan monthly payment).
5. Be creative! Use colors, drawings, or digital design to make your work clear and engaging.

Option 3: Song / Jingle / Rap

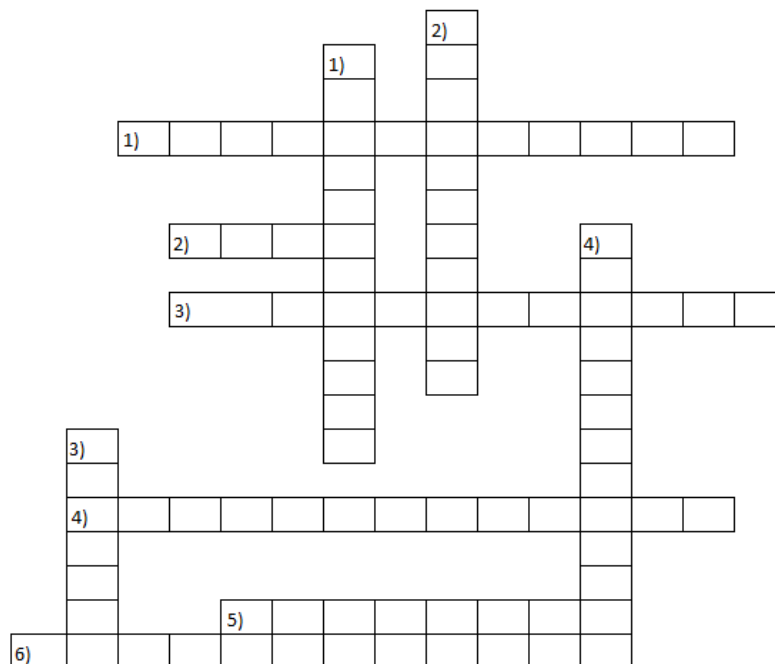
1. Form small groups (5–6 members).
2. Compose a short jingle, rap, or song (1–2 minutes) that explains:
 - The differences between Business Loans and Consumer Loans.
 - Key words like *purpose*, *repayment terms*, *requirements*, *interest rates*.
3. Include at least one line that shows a loan computation (e.g., monthly payment).
4. Perform live or record your performance to present to the class.

Option 4: Storyboard / Comic Strip

1. Form small groups (5-6 members)
2. Create a 4–6 panel comic strip or storyboard showing a character applying for a loan.
3. Your comic must include:
 - The character's reason for borrowing.
 - The bank officer explains whether it is a Business or Consumer Loan.
 - The loan's requirements, repayment terms, and interest rate.
 - At least one sample loan computation (mortgage or amortization).
4. Submit your comic strip on short bond paper or digitally.

V. Synthesis:	Compare the key differences between Business Loans and Consumer Loans using examples from your group activity. Which loan type is more suitable for your character's situation and why? Use at least two characteristics (e.g., purpose, interest rate, repayment terms) to support your answer. Based on the loan computation shown in your activity, evaluate whether the borrower can afford the loan. Should the borrower proceed with the loan or choose another option? Explain your decision using financial reasoning (e.g., monthly payment vs income, total interest, loan term). If the borrower fails to pay the loan on time, what are the possible consequences? How can borrowers avoid these problems? Use your activity's scenario to explain at least two possible outcomes (e.g., penalties, loss of collateral, damaged credit).
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I. Activity No. 3	Assessing Learning Outcomes
II. Objectives:	At the end of the activity, you should be able to solve problems involving solve problems involving business and consumer loans.
III. Materials Needed:	pen and paper
IV. Instructions:	Perform the indicated task. Write your answer on a separate sheet of paper. A. Loan Identification. Write B if the statement describes a <i>Business Loan</i> and C if it describes a <i>Consumer Loan</i>. - Borrowed mainly to start, maintain, or expand a business. - Typically fixed repayment with monthly installments over a set period. - Requires business plans, financial records, and collateral. - Borrowed for personal or household use such as education or appliances. - Interest rates can be lower since the loan is secured and intended to generate income. - Requirements are generally simpler and may only include proof of income or employment. - Repayment terms may be flexible and based on income. - Often has higher interest rates because it is unsecured and depends on personal income. - Intended for business purposes like purchasing raw materials or equipment. - Used for personal consumption, such as buying a car or paying tuition. - Ana borrowed ₱200,000 to buy baking equipment for her pastry shop. - Carlo applied for ₱50,000 to buy a new motorcycle for personal use. - A company borrowed ₱1,500,000 to open another branch in Cebu. - Maria borrowed ₱30,000 to pay for her child's tuition fee. - A tricycle operator borrowed ₱100,000 to buy a second-hand unit for public transport service. B. Crossword Puzzle. Complete the crossword puzzle by filling in the correct loan-related terms based on the given clues.



Across

1. A _____ is money lent specifically for business purposes, such as starting or expanding a business.
2. The _____ of loan refers to the length of time given to fully pay a loan.
3. The _____ is the method of paying a loan in equal installments covering both principal and interest.
4. The _____ for consumer loans are often fixed, requiring monthly installments over a set period.
5. A _____ is a loan secured by collateral that must be repaid according to agreed terms.
6. Business loans often require documents such as business plans, collateral, and credit history; these are called _____.

Down

1. A _____ is money lent to an individual for personal or family use, such as buying a house or car.
2. The _____ balance is the amount of debt that still remains unpaid at a specific time.
3. The _____ of a business loan is usually to fund capital, equipment, or raw materials.
4. Consumer loans usually have higher _____ than business loans because they are often unsecured.

C. **Problem Solving.** Solve the following problems correctly

1. A loan of P200,000 is to be repaid in full 3 years. If the interest rate is 8% per annum. How much will be paid after 3 years?

	2. For a purchase of a house and lot worth P3,800,000 the bank requires 20% down payment, find the mortgaged amount. 3. A car dealer offers a 15% down payment for the purchase of the car. How Much is the mortgage amount if the cash value of the car is P1,500,000? 4. Mr. Ramos is considering paying his outstanding balance after 3 years of payment. The original amount of the loan is P100,000 payable annually in 5 years. If the interest rate is 10% per annum and the regular payment is P26,379.75 annually, how much is the outstanding balance after the 3rd payment? 5. A Consumer loan worth P30,000 is to be repaid in 12 months at 9% convertible monthly: How much is the monthly payment?
V. Synthesis:	1. Based on the loan characteristics in Section A and the crossword terms in Section B, explain how the purpose of a loan affects its repayment terms and requirements. Use at least two examples from Section A to support your answer. 2. Choose one problem from Section C. Explain whether the loan involved is a business loan or a consumer loan. Then, describe if the loan were the other type (business vs consumer). 3. From Section C, pick the scenario that you think is most financially risky. Explain why it is risky using at least two loan concepts (e.g., interest rate, term, outstanding balance, monthly payment). Then, suggest one strategy to reduce the risk (e.g., increase down payment, shorten loan term, choose lower interest rate).

Objective: By the end of this lesson, learners are able to create a financial proposal using comparisons of different investment plans to recommend and explain the best financial strategy.

Title: Building a Financial Proposal Through Smart Investing

Groupings:

- Learners will work in groups of 4–6 members.
- Each member has a specific role:
 1. Researcher (given data handler): Uses the provided investment data (interest rates, terms, etc.) as inputs.
 2. Mathematician/Analyst: Computes future value, present value, and interest based on the given data.
 3. Data Manager: Creates tables, graphs, and charts.
 4. Writer/Editor: Drafts the financial proposal with explanations.
 5. Presenter(s): Lead the delivery of the group presentation.

Task Description:

As a team of financial consultants, you are tasked to design a financial proposal for a given client profile (see sample cases below). You will compare three investment plans using the provided investment data and recommend the best financial strategy.

Sample Investment Data (Provided by Teacher)

(These are only examples; teacher may adjust as needed)

- Time Deposit: 3.5% annual interest, compounded annually, minimum ₱50,000, term 5 years
- Mutual Fund: Average return of 7% annually, compounded quarterly, moderate risk
- Pension Plan: 5% guaranteed annual growth, compounded annually, long-term (10+ years)
- Stocks (optional): 10% average annual growth, high risk, compounded annually

Sample Client Profiles (Choose or Assign)

- Rosa, age 25, wants to start saving for her retirement at age 60.
- A young couple, both age 30, are saving for their child's college education in 15 years.
- Mark, age 40, wants to invest for a business expansion fund in 7 years.

Process & Output:

1. Use the given investment data (no external research needed).
2. Compute maturity values, present values, and growth using simple/compound interest formulas.
3. Compare risks, returns, and suitability for the client.
4. Create a Financial Proposal (written report or digital presentation).
5. Present findings as a group (10–15 minutes).

Rubrics

Criteria	4 (excellent)	3 (Proficient)	2 (Developing)	1 (Beginning)
Mathematical Accuracy (Group)	All computations are correct.	A few minor errors.	Some errors affecting results.	Many errors
Comparative Analysis (Group)	In-depth, insightful, well-organized.	Adequate comparison, some insights.	Limited depth.	Incomplete or unclear.
Recommendation (Group)	Clear, realistic, fully supported.	Mostly supported, somewhat realistic.	Weak justification	No clear recommendation.
Collaboration & Role Fulfillment (Individual)	Fully contributed, role well-executed.	Contributed with minor lapses.	Uneven contribution.	Minimal or no contribution.
Presentation (Group)	Clear, creative, professional	Organized, mostly clear.	Somewhat unclear.	Disorganized, hard to follow.